AIR5011 / MDS5122 / MBI6011
Assignment 1

Due: 23:59, March 11, 2026

Instructions:

1. You must submit your assignment on Blackboard. Please upload a PDF file along with your code. Upload the PDF file and the code (as a zip file) separately on Blackboard, and do not include the report inside the zip file.
2. Your submission needs to be clearly-structured and well-presented to receive full credit. Please detail how you leverage your acquired knowledge to solve the problem and present your proposed solution step by step. Ensure that your solutions are easy to follow and written in English.

A. Build a Neural Network Using PyTorch (40 points)

To implement a deep learning application, the simplest approach is to write the code using a modern deep learning framework. The most popular framework in academia is PyTorch. In the first part, we will develop an image classification application using a toy dataset, Fashion-MNIST (60,000 training samples and 10,000 test samples, each being a 28x28 grayscale image), and experiment with various factors to enhance performance on the testing dataset. Your code must be written using PyTorch (with version 2.0 or higher) and Jupyter Notebooks. The notebooks should include your experimental results and necessary comments.

You may experiment with different network designs (such as residual connections and stacks of 3x3 convolutions), components (such as depthwise separable convolutions), optimizers (such as AdamW and Muon), training schemes (such as warmup and linear learning rate schedulers), or any other factors you can think of to increase accuracy; you may design your own architecture with components such as residual connections, but do not import or replicate standard models like ResNet-18 from torchvision or other libraries. We highly encourage you to explore your own designed solution.

The accuracy is evaluated on the test set and should be as high as possible. However, we do not score your assignment solely based on the achieved accuracy; instead, we seek evidence of your effort and multiple attempts to improve accuracy incrementally in your experimental report. You must detail the steps taken to enhance accuracy, including the factors you have considered and evaluated, and explain why you chose to incorporate each factor. Additionally, you should include the following sections in your submission to receive the corresponding credits.

Scoring Criteria:

1. Set up the Python coding environment and write a simple CNN as shown below. Train it on the Fashion-MNIST dataset for 10 epochs, and report the accuracy; this accuracy will serve as the baseline accuracy. **(10 points)**

```
model = nn.Sequential(
    nn.Conv2d(1, 256, 5, padding=2), nn.ReLU(),
    nn.Conv2d(256, 256, 5, padding=2), nn.ReLU(),
    nn.Conv2d(256, 256, 5, padding=2), nn.ReLU(),
    nn.Conv2d(256, 256, 5, padding=2), nn.ReLU(),
    nn.AdaptiveAvgPool2d(1),
    nn.Flatten(),
    nn.Linear(256, 10)
)
```

2. What factors did you consider to improve accuracy? For example, you might try using a deeper and wider convolutional neural network, a better optimizer, data augmentation methods, dropout layers, or a longer training duration. These are just a few suggestions; you may explore other options as well. We expect your final accuracy to be higher than the baseline accuracy. **(20 points)**
3. Your code and submitted document are expected to be clean and well-organized. For example, the paragraphs are well organized; the results are presented in tables and effectively visualized through high-quality figures; it would be better if the documents are written in LaTeX. **(5 points)**
4. What have you learned through your attempts in the assignment? **(5 points)**

B. Build a Neural Network From Scratch (60 points)

To gain a better understanding of what a neural network is, you are required to write, train, and evaluate your own deep neural networks without using any deep learning libraries. You are prohibited from using any deep learning frameworks or auto-differentiable libraries, such as PyTorch, TensorFlow, and JAX; instead, you should use only NumPy to implement the functionalities. If you find performance to be suboptimal, you should implement these functions with vectorization to enhance speed.

Notice:

1. You should implement both the forward and backward passes for each network module, ensuring they are similar to how PyTorch implements them; for example, the module should be implemented as a Python class.
2. Some network modules shown below are marked with *config*, indicating that they should be initialized with these parameters during their setup. Please refer to PyTorch's documentation for the meanings of these names. Your implementations should closely resemble those of PyTorch.

Scoring Criteria:

1. Implement the following network modules: **(40 points)**
   a) GELU, use tanh approximation **(5 points)**

   $$\mathrm{GELU}(x) \approx 0.5x\left(1+\tanh\left(\sqrt{\frac{2}{\pi}}(x+0.044715x^3)\right)\right)$$

   b) Linear, with config: *in_features, out_features, bias* **(5 points)**
   c) CrossEntropyLoss **(5 points)**
   d) Adam, with config: *lr* **(5 points)**
   e) Conv2d, with config: *in_channels, out_channels, kernel_size, stride, bias* **(10 points)**
   You MUST NOT use any pre-existing convolution functions
   f) SwitchNorm2d, refer to the paper (arXiv:1907.10473), with config: *num_features* **(10 points)**
2. Using the modules you created, develop a simple CNN to train the network on the Fashion-MNIST dataset. Compare its performance with that of a PyTorch counterpart to assess whether the accuracy is similar. **(10 points)**

3. Your code and submitted document are expected to be clean and well-organized. For example, the paragraphs are well organized; the results are presented in tables and effectively visualized through high-quality figures; it would be better if the documents are written in LaTeX. **(5 points)**
4. What have you learned through your attempts in the assignment? **(5 points)**

Environment Setup Guide

1) Install Miniforge
   - ✓ Go to https://github.com/conda-forge/miniforge?tab=readme-ov-file#install
   - ✓ Follow the instructions to install it on your system.
2) Open Command Prompt
   - ✓ Press Windows + R, type *cmd*, press Enter
3) Create a Python environment
   - ✓ Type: *conda create -n myenv python=3.10 -y*
   - ✓ Then: *conda activate myenv*
4) Install PyTorch
   - ✓ For CPU only (works on all computers):
     - ■ *pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cpu*
   - ✓ For NVIDIA GPU (only if you have one):
     - ■ Go to https://pytorch.org, choose your CUDA version, and copy the pip command
5) Install extra tools
   - ✓ Type: *pip install matplotlib ipykernel*
6) Install VS Code
   - ✓ Download from https://code.visualstudio.com/
   - ✓ Open VS Code, install the "Python" extension (by Microsoft)
   - ✓ Open a .py file and select the Python interpreter (myenv) located in the bottom right corner.
7) Test it
   - ✓ Create a file test.py with:

```
import torch
print(torch.__version__)
```

   - ✓ Run it in VS Code or in Command Prompt (with "myenv" activated). You should see a version number like 2.5.0.

If you encounter any issues, please do not hesitate to ask in the Class WeChat group.